Chapter 4 solutions

CH04-E1

1. Solve

$$x_1 + 2x_2 = 0, \qquad x_2 + x_3 = 0.$$

Set $x_2 = t$, giving $x_1 = -2t, x_3 = -t$. Hence

$$N(A) = \operatorname{span}\{(-2, 1, -1)^T\}.$$

2. The independent rows $r_1 = (1, 2, 0)^T, r_2 = (0, 1, 1)^T$ form a row-space basis. For $n = (-2, 1, -1)^T$,

$$r_1^T n = -2 + 2 = 0, \qquad r_2^T n = 1 - 1 = 0.$$

Every row-space combination is therefore perpendicular to n. The orthogonal spaces have dimensions 2 and 1, summing to 3, so they are orthogonal complements.

3. The 2×3 matrix has two independent rows and rank 2. Its column space is a two-dimensional subspace of $\mathbb{R}^2$, hence all of $\mathbb{R}^2$. The left nullity is

$$\dim N(A^T) = m - r = 2 - 2 = 0,$$

Thus $N(A^T) = \{0\}$.

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CH04-E2

1. Since $a^T a = 3$ and $a^T b = 7$,

$$p = \frac{7}{3} a = \frac{7}{3}(1, 1, 1)^T.$$

2.

$$e = b - p = \left(2 - \frac{7}{3}, 1 - \frac{7}{3}, 4 - \frac{7}{3}\right)^T = \frac{1}{3}(-1, -4, 5)^T.$$

3. $a^T e = (-1 - 4 + 5)/3 = 0$, proving orthogonality to the line.

4. The line projector is

$$P = \frac{a a^T}{a^T a} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

It is symmetric. The product $J_3 J_3$ has every entry equal to 3, so

$$P^2 = \frac{1}{3^2} \cdot 3\, J_3 = \frac{1}{3} J_3 = P,$$

where J_3 is the all-ones matrix.

CH04-E3

1.

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, \quad x = \begin{bmatrix} c \\ m \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}.$$

2.

$$A^T A = \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix}, \qquad A^T b = \begin{bmatrix} 5 \\ 6 \end{bmatrix}.$$

The equations are

$$3c + 3m = 5, \qquad 3c + 5m = 6.$$

Subtraction gives $2m = 1$, hence $m = 1/2$, and substitution gives $c = 7/6$.

3.

$$p = Ax = \left(\frac{7}{6}, \frac{7}{6} + \frac{3}{6}, \frac{7}{6} + \frac{6}{6}\right)^T = \left(\frac{7}{6}, \frac{5}{3}, \frac{13}{6}\right)^T.$$

$$e = b - p = \left(-\frac{1}{6}, \frac{1}{3}, -\frac{1}{6}\right)^T.$$

The residual sum is zero, establishing orthogonality to the constant column. Also

$$t \cdot e = 0 + \frac{1}{3} + \left(-\frac{1}{3}\right) = 0.$$

Hence $A^T e = 0$.

4. $\|e\|^2 = 2/36 + 4/36 = 1/6$, the minimum sum of squared residuals over all c, m.

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CH04-E4

First verify invertibility: column independence gives $z^T A^T A z = \|Az\|^2 > 0$ for $z \neq 0$, so $A^T A$ is invertible.

1. Since $A^T A$ is symmetric, its inverse is symmetric.

$$P^T = A((A^T A)^{-1})^T A^T = A(A^T A)^{-1} A^T = P.$$

Also

$$P^2 = A(A^T A)^{-1} A^T A(A^T A)^{-1} A^T = P.$$

2. The complement is symmetric and

$$(I - P)^2 = I - 2P + P^2 = I - P.$$

Thus it is an orthogonal projector. The cross product is

$$P(I - P) = P - P^2 = 0.$$

Its range is perpendicular to the range of P because $P^T(I - P) = 0$.

3. If $A^T A z = 0$, multiply by z^T to get

$$0 = z^T A^T A z = \|Az\|^2.$$

Therefore $Az = 0$. Column independence gives $z = 0$, proving a trivial nullspace and invertibility of $A^T A$.

4. Dependent columns give nonzero z with $Az = 0$, hence

$$A^T A z = 0,$$

This nonzero nullspace vector proves singularity of $A^T A$.

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CH04-E5

1. The norm of a_1 is $\sqrt{2}$, so

$$q_1 = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right)^T.$$

Since $q_1^T a_2 = 1/\sqrt{2}$,

$$z_2 = a_2 - \frac{1}{2}(1,1,0)^T = \left(\frac{1}{2}, -\frac{1}{2}, 1\right)^T.$$

$$\|z_2\| = \sqrt{\frac{1}{4} + \frac{1}{4} + 1} = \sqrt{\frac{3}{2}},$$

$$q_2 = \frac{1}{\sqrt{6}}(1,-1,2)^T.$$

2. The Gram-Schmidt representations are

$$a_1 = \|a_1\| q_1 = \sqrt{2}\, q_1,$$

$$a_2 = \frac{1}{\sqrt{2}} q_1 + \sqrt{\frac{3}{2}} q_2.$$

Therefore

$$R = \begin{bmatrix} \sqrt{2} & 1/\sqrt{2} \\ 0 & \sqrt{3/2} \end{bmatrix}.$$

3. Both columns have unit norm, and

$$q_1^T q_2 = \frac{1 + (-1) + 0}{\sqrt{12}} = 0,$$

Thus $Q^T Q = I_2$. Expanding by columns gives

$$\sqrt{2} q_1 = a_1,$$

$$\frac{1}{\sqrt{2}} q_1 + \sqrt{\frac{3}{2}} q_2 = \left(\frac{1}{2}, \frac{1}{2}, 0\right)^T + \left(\frac{1}{2}, -\frac{1}{2}, 1\right)^T = a_2,$$

Hence $QR = A$.

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CH04-E6

1.

$$Q^T b = \left(\frac{1+2}{\sqrt{2}}, \frac{1-2}{\sqrt{6}}\right)^T = \left(\frac{3}{\sqrt{2}}, -\frac{1}{\sqrt{6}}\right)^T.$$

Solve

$$\sqrt{2}\,x_1 + \frac{1}{\sqrt{2}}x_2 = \frac{3}{\sqrt{2}}, \qquad \sqrt{\frac{3}{2}}x_2 = -\frac{1}{\sqrt{6}}.$$

The second equation gives $x_2 = -1/3$; the first then gives $x_1 = 5/3$. Thus

$$x = \begin{bmatrix} 5/3 \\ -1/3 \end{bmatrix}.$$

2.

$$p = Ax = \left(\frac{4}{3}, \frac{5}{3}, -\frac{1}{3}\right)^T,$$

$$e = b - p = \frac{1}{3}(-1, 1, 1)^T, \qquad \|e\|^2 = \frac{1}{3}.$$

3.

$$a_1^T e = -\frac{1}{3} + \frac{1}{3} = 0,$$

$$a_2^T e = -\frac{1}{3} + \frac{1}{3} = 0.$$

This nonzero residual distinguishes a least-squares solution from an exact solution. Columnwise orthogonality is precisely $A^T e = 0$, and full column rank guarantees a unique parameter vector.

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CH04-E7

1. The columns are identical, so they are dependent and rank is 1.
2. Set $a = (1, 2)^T$. Then
$$a^T a = 5, \qquad a^T b = 1 + 6 = 7,$$
$$p = \frac{7}{5} a = \left(\frac{7}{5}, \frac{14}{5} \right)^T.$$
3.
$$A^T A = \begin{bmatrix} 5 & 5 \\ 5 & 5 \end{bmatrix}, \qquad A^T b = \begin{bmatrix} 7 \\ 7 \end{bmatrix}.$$

Every minimizer has the form
$$x^\star = \begin{bmatrix} 7/5 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \quad t \in \mathbb{R}.$$

4. Differences of parameters producing p lie in $N(A) = \text{span}\{(-1, 1)^T\}$. This nonzero direction gives nonunique parameters, while all map to the same nearest column-space point p.

CH04-E8

1. Direct multiplication gives
$$P^2 = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} = P.$$
2. However,
$$P^T = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} \neq P.$$
3. Since $P(x, y)^T = (x + y, 0)^T$,
$$R(P) = \text{span}\{(1, 0)^T\}, \qquad N(P) = \text{span}\{(-1, 1)^T\}.$$
4. $R(P)$ and $N(P)$ are not orthogonal, since $(1, 0)^T \cdot (-1, 1)^T = -1 \neq 0$. Thus $P^2 = P$ describes an oblique projector; orthogonal projection also requires $P^T = P$.

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CH04-E9

The row space has basis $(1,1,0)^T$, the column space $(1,2)^T$, both of dimension 1. Solving Ax=0 gives basis $(1,-1,0)^T, (0,0,1)^T$, dimension 2. The left nullspace has basis $(2,-1)^T$, dimension 1. Their dot products verify the two orthogonal pairs, and dimensions sum to 3 on the input side and 2 on the output side.

Projection onto c=(1,2) gives $p = c(c^Tb)/(c^Tc) = (1,2)^T$, $e = (2,-1)^T$, with $c^Te=0$. For any x, e is orthogonal to p−Ax, so

$$\|b - Ax\|^2 = 5 + \|p - Ax\|^2 \geq 5.$$

Equality holds exactly when Ax=p, or $x_1+x_2=1$. Hence all minimizers are $x = (1/2 + t, 1/2 - t, s)^T$. Split this into $\hat{x} = (1/2, 1/2, 0)^T$ and $z = (t, -t, s)^T$, in the orthogonal row and null spaces. Thus $\|x\|^2 = 1/2 + 2t^2 + s^2$, uniquely minimized at t=s=0, with norm $1/\sqrt{2}$.

For the particular solution, $(2,-1,2)^T = \hat{x} + (3/2, -3/2, 2)^T$. Its nullspace component maps to zero and has squared norm 17/2. Its total squared norm is 1/2+17/2=9, so its norm is 3. It has the same projected output p and minimum residual norm $\sqrt{5}$, but a larger parameter norm. Residual norm and parameter norm are distinct objectives.

CH04-E10

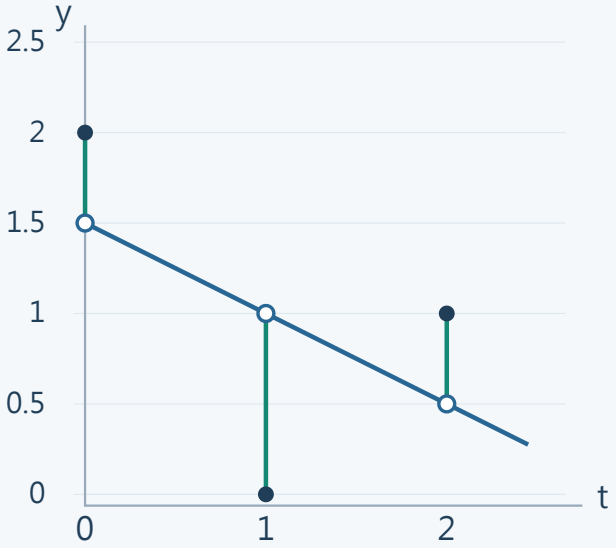
The design matrix remains $A = [(1,1,1)^T\ (0,1,2)^T]$, but $b = (2,0,1)^T$. The normal equations are $3c + 3m = 3$, $3c + 5m = 2$. Subtraction gives $2m = -1$, hence $m = -1/2, c = 3/2$.

Time t	Observed	Predicted	Residual	Squared residual
0	2	$3/2$	$1/2$	$1/4$
1	0	1	-1	1
2	1	$1/2$	$1/2$	$1/4$

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CH04-E10 (continued)

Hold t fixed: measure three vertical residuals



Filled dots: observations b_i

Open dots: predictions p_i

Blue line: $y = 3/2 - t/2$

Green: $e_i = b_i - p_i$

$e = (1/2, -1, 1/2)^T$

$\Sigma e_i^2 = 3/2$

$1/2 - 1 + 1/2 = 0$

$0(1/2) + 1(-1) + 2(1/2) = 0$

Equal units on t and y; every green segment holds t fixed.

The residual is $e = (1/2, -1, 1/2)^T$, with $\|e\|^2 = 3/2$. The checks $1/2 - 1 + 1/2 = 0$ and $-1 + 2(1/2) = 0$ establish $A^T e = 0$.

Since $A^T e = 0$, every $h = (\Delta c, \Delta m)^T$ satisfies

$$\|b - A(x^\star + h)\|^2 = \frac{3}{2} + \|Ah\|^2 = \frac{3}{2} + 3(\Delta c + \Delta m)^2 + 2(\Delta m)^2.$$

The increment vanishes only at $h = 0$, proving uniqueness. A zero sum checks only the first column: $d = (1, -1, 0)^T$ sums to zero but has inner product -1 with the time column.

After centering, the orthogonal columns have right-hand-side inner products 3 and −1. Hence $a = 3/3 = 1, m = -1/2, c = a - m = 3/2$. Thin QR uses the same Q, R as the lesson, but now

$$Q^T b = \begin{bmatrix} \sqrt{3} \\ -1/\sqrt{2} \end{bmatrix}, \qquad q_3^T b = \frac{3}{\sqrt{6}}, \qquad (q_3^T b)^2 = \frac{3}{2}.$$

The triangular equations give $\sqrt{2}m = -1/\sqrt{2}$ and $\sqrt{3}(c + m) = \sqrt{3}$, confirming the normal-equation result. Plot segments hold time fixed and compare heights. After combining observations into a three-dimensional vector, e is perpendicular to the entire reachable prediction space $C(A)$. The two drawings have different coordinate spaces and different objects to which perpendicularity refers.

CH04-E11

Write $p(x) = a + bx$. Orthogonality gives

$$a + \frac{b}{2} = \frac{1}{3}, \qquad \frac{a}{2} + \frac{b}{3} = \frac{1}{4}.$$

Thus $a = -\frac{1}{6}, b = 1, p(x) = x - \frac{1}{6}$, and the residual is $r = x^2 - x + \frac{1}{6}$.

Direct integration gives $\int_0^1 r(x)\,dx = 0$ and $\int_0^1 xr(x)\,dx = 0$, so the residual is orthogonal to the entire subspace. Expanding,

$$\int_0^1 r(x)^2 dx = \int_0^1 \left(x^4 - 2x^3 + \tfrac{4}{3}x^2 - \tfrac{1}{3}x + \tfrac{1}{36}\right) dx = \frac{1}{180}.$$

This is the uniquely minimized integrated squared error; its square root is the residual norm induced by the inner product.

References and attribution

Lin Cing, handwritten linear algebra notes. Gilbert Strang, Introduction to Linear Algebra, 5th ed., Chapter 4.

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